Amicable Numbers

Definition:

Two different positive integers x and y form an amicable pair if,

(i) The sum of proper divisors of x is y.

(ii) The sum of proper divisors of y is x.

A poper divisor is a positive divisor other than the number itself.

Example: 220 and 284 are amicable because $s(220) = 284$ and $s(284) = 220$

Algorithm:

(1) PreCompute Proper-divisor sums up to MAX_N

(2) Read pairs until 0 0 is entered.

(3) For each i in $[a, b]$, let $j = sumDiv[i]$

(4) Count the pair only if $j > i$, $a <= j <= b$, and $sumDiv[j] == i$,

(5) Print the count.

Complexity:

Precomputation: $O(N \log N)$

Each test case: $O(b-a+1)$

space: $O(N)$.

Precomputed divisor sums make each test case efficient. The condition $j > i$ Prevents counting the same pair twice.

Code:

```
#include <bits/stdc++.h>
using namespace std;

const int MAXN=100000;
int sumDiv[MAXN+1];
int main()
{
    int i, j;
```

```
for (i=1 ; i<=MAXN/2 ; i++){
        for(j=2*i; j<=MAXN; j+=i){
            sumDiv[j]+=i;
        }
    }
int a, b;
while (cin>>a>>b && (a!=0 || b!=0)){
    int ans=0;
    for(int i=max(1,a); i<=b; i++){
        int j=sumDiv[i];
        if (j>i && j>=a && j<=b && sumDiv[j]
            ==i){
                ans++;
            }
        }
    cout<<ans<<'\n';
}
return 0;
}.
```